

Wei Liang, P.E., Ph.D.

📍 8 Architecture Dr, #03-02, Singapore 117564 • 📞 +65 8743 4325

✉️ wliang@nus.edu.sg • 🔗 <https://linkedin.com/in/wadeliang/>

🌐 <https://lw-g38.github.io/> • 📠 +1 201-744-1353

Research interests

Artificial intelligence for buildings, Data-driven modeling and controls, Robotics for building management, IoT and Smart Buildings, Indoor Environmental Quality

Education

- 2019 – 2024 **Carnegie Mellon University (CMU)** – Pittsburgh, PA, USA
Ph.D. in Building Performance and Diagnostics
Advisors: Prof. Erica Cochran Hameen & Prof. Vivian Loftness
- 2012 – 2014 **University of California, Merced (UC Merced)** – Merced, CA, USA
Master of Science in Mechanical Engineering
Advisor: Prof. Jian-Qiao Sun
- 2008 – 2012 **Nanjing University (NJU)** – Nanjing, Jiangsu, China
Bachelor of Science in Acoustics

Research experience

- Jul 2024 – Present **Senior Research Fellow, National University of Singapore (NUS)**
Mentors: Prof. Adrian Chong
- Authored major portions of successful proposals to secure funding from multiple funding agencies (NUS, A*STAR & GBIC) over SG\$3.2M
 - Led research on comparative evaluations of LLM, RL, and rule-based building controls on BOPTEST
 - Led interdisciplinary research projects, including overseas collaborations, coordination with industrial partners, and supervising Ph.D. students
 - Designed LoRaWAN-based ecosystems for occupancy counting, occupancy-centric control, and mixed-mode cooling systems for multiple projects across office buildings, schools, and hospitals
 - Real-world implementation of optimal controls on building automation systems
 - Adversarial inverse reinforcement learning for complex building HVAC systems

Aug 2019 – June 2024	Research Assistant, CMU Advisors: Prof. Erica Cochran Hameen & Prof. Vivian Loftness • Mobile sensing robot for indoor environmental quality assessments • Spatiotemporal Gaussian Process modeling for indoor environmental quality • Spatial mapping of thermal comfort with vSLAM and semantic segmentation • Social robot for post-occupancy evaluation • Iterative learning controls for building thermodynamics
May 2023 – Aug 2023	Ph.D. Researcher, Lawrence Berkeley National Lab (LBNL) Advisors: Dr. Tianzhen Hong • Physics-informed neural network (PINN) modeling for building thermodynamics • Model Predictive Control (MPC) to optimize building energy flexibility
Oct 2022 – May 2023	Ph.D. Researcher, Pacific Northwest National Lab (PNNL) Advisors: Dr. Michael Brambley • Created a gradient-boosting-based framework for long-term energy consumption prediction for packaged air conditioning units with limited information
Aug 2012 – Jun 2015	Research Assistant, UCM Advisors: Prof. Jian-Qiao Sun • ARMAX modeling for building dynamics and MPC optimal control for AHUs
Teaching experience	
Jan 2022 – May 2024	Instructor, 48722, Building Performance Modeling, CMU Sole instructor of a graduate-level class • Teaching building simulations and Building Physics • Independently designed teaching materials and lab assignments • 15-30 enrollments with diverse academic and professional backgrounds
Fall 2020 – Fall 2021	Teaching Assistant, Multiple courses, CMU 48768, Indoor Environmental Quality, Fall 2021 48729, Sustainability, Health and Productivity, Fall 2021 48722, Building Performance Modeling, Spring 2021 48721, Building Controls and Diagnostics, Spring 2021 48116, Building Physics, Fall 2020
Spring 2013 – Spring 2014	Lab Instructor and Teaching Assistant, Multiple courses, UC Merced ME 021, Engineering Computing, Fall 2014, Summer 2014 & Fall 2013 ME 140, Vibration and Control, Spring 2014 & Spring 2013
Summer 2013 – Spring 2014	Teaching Assistant, Multiple courses, UC Merced ME 142, Mechatronics, Spring 2014 ENGR 065, Circuit Theory, Summer 2013

Industry experience

- Aug 2015 – **Mechanical Engineer (P.E.), Gayner Engineers** – San Francisco, CA, USA
July 2019
- HVAC System and Building Automation System design for critical buildings
 - Building energy simulations for green building rating systems
 - Commissioning and Energy Auditing for MEP and medical equipment systems
 - Project management, construction administration, and drawing productions

Journal Publications

- J0 **Liang, W., & Chong, A.** (2026). Adversarial inverse reinforcement learning control for mixed-mode ventilation systems with physics-constrained neural network [Under review]. *Energy and Buildings*
- J1 **Liang, W., Zhang, Y., Chong, A., Hameen, E. C., & Loftness, V.** (2025). Exploring gaussian process regression for indoor environmental quality: Spatiotemporal thermal and air quality modeling with mobile sensing. *Building and Environment*, 281, 113143. <https://doi.org/10.1016/j.buildenv.2025.113143>
- J2 **Liang, W., Li, H., Zhan, S., Chong, A., & Hong, T.** (2024). Energy flexibility quantification of a tropical net-zero office building using physically consistent neural network-based model predictive control. *Advances in Applied Energy*, 14, 100167. <https://doi.org/10.1016/j.adapen.2024.100167>
- J3 **Liang, W., Ma, S., Cochran, E., & Flanigan, K. A.** (2023). Distributed MPC-ILC thermal control design for large-scale multi-zone building hvac system. *SIGENERGY Energy Inform. Rev.*, 3(2), 34–46. <https://doi.org/10.1145/3607114.3607118>
- J4 **Liang, W., Quinte, R., Jia, X., & Sun, J.-Q.** (2015). Mpc control for improving energy efficiency of a building air handler for multi-zone vavs. *Building and Environment*, 92, 256–268. <https://doi.org/10.1016/j.buildenv.2015.04.033>
- J5 Min Ye, Y., **Liang, W.**, Xu, F., Chong, A., Barnes, C. M., & Yam, K. C. (2026). The effects of mixed-mode ventilation on energy saving and employee job satisfaction, work engagement, and job performance. *Scientific Reports*. <https://doi.org/https://doi.org/10.1038/s41598-026-38812-0>
- J6 Cho, J., **Liang, W.**, Cheng, S., Chong, A., & Heo, Y. (2026). Weather-adaptive rule-based mixed-mode ventilation in a tropical climate: Simulation and real-world validation. *Energy and Buildings*, 352, 116816. <https://doi.org/https://doi.org/10.1016/j.enbuild.2025.116816>
- J7 Wang, X., Jiang, Z., Zhan, S., **Liang, W.**, & Dong, B. (2026). Dual-phase evaluation framework of physics-informed machine learning model for building dynamics modeling and advanced control. *Energy and Buildings*, 352, 116791. <https://doi.org/https://doi.org/10.1016/j.enbuild.2025.116791>

- J8 Qaisar, I., **Liang**, W., Sun, K., Xing, T., & Zhao, Q. (2024). An experimental comparative study of energy saving based on occupancy-centric control in smart buildings. *Building and Environment*, 268, 112322. <https://doi.org/10.1016/j.buildenv.2024.112322>
- J9 Xiong, R., Shi, Y., Jing, H., **Liang**, W., Nakahira, Y., & Tang, P. (2024). Calibrating subjective data biases and model predictive uncertainties in machine learning-based thermal perception predictions. *Building and Environment*, 247, 111053. <https://doi.org/10.1016/j.buildenv.2023.111053>
- J10 Xiong, F., Qin, Z., Hernández, C., Sardahi, Y., Narajani, Y., **Liang**, W., Xue, Y., Schütze, O., & Sun, J. (2013). A multi-objective optimal pid control for a nonlinear system with time delay. *Theoretical and Applied Mechanics Letters*, 3(6), 063006. <https://doi.org/10.1063/2.1306306>
- J11 Hernández, C., Naranjani, Y., Sardahi, Y., **Liang**, W., Schütze, O., & Sun, J.-Q. (2013). Simple cell mapping method for multi-objective optimal feedback control design. *International Journal of Dynamics and Control*, 1(3), 231–238. <https://doi.org/10.1007/s40435-013-0021-1>

Peer-reviewed Conference Publications

- C1 **Liang**, W., & Chong, A. (2025). Learning adaptive mixed-mode ventilation policies via adversarial inverse reinforcement learning. *ICML 2025 CO-BUILD Workshop on Computational Optimization of Buildings*
- c2 Zhan, S., **Liang**, W., & Chong, A. (2025). Evaluating the practical energy saving potential of a personal cooling system in the tropics. *Proceedings of Building Simulation 2025: 19th Conference of IBPSA*, 19. <https://doi.org/10.26868/25222708.2025.1845>
- C3 **Liang**, W., Zhang, Y., Zhang, J., & Hameen, E. C. (2024). An expeditious spatial mean radiant temperature mapping framework using visual slam and semantic segmentation [In 2024 CVPR Computer Vision in the Built Environment Workshop]. <https://doi.org/10.48550/arXiv.2410.09443>
- C4 **Liang**, W., & Brambley, M. R. (2024). Event-based energy impact tracking and forecasting with limited measurements for rooftop units. *ASHRAE Transactions*, 239–248. <https://doi.org/10.63044/s24lia26>
- C5 **Liang**, W., Ma, S., Hameen, E. C., & Flanigan, K. (2022). Integrated MPC-ILC control design for thermal control of a large-scale multi-zone building hvac system. *Proceedings of the 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, 129–138. <https://doi.org/10.1145/3563357.3564068>
- C6 **Liang**, W., Xiong, R., Liu, P., Tang, P., & Hameen, E. C. (2022). Improving post-occupancy evaluation engagement using social robots. *Proceedings of the 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, 159–167. <https://doi.org/10.1145/3563357.3564071>
- C7 Kumar, E., Hameen, E. C., & **Liang**, W. (2020). Global marginal carbon footprint evaluation of internet services with building energy models. *BuildSIM-Nordic 2020*

- C8 Sardahi, Y., Naranjani, Y., **Liang**, W., Sun, J.-Q., Hernandez, C., & Schuetze, O. (2013). Multi-objective optimal control design with the simple cell mapping method. *ASME 2013 International Mechanical Engineering Congress and Exposition*, V04BT04A025. <https://doi.org/10.1007/s40435-013-0021-1>

Research Funding

- Oct.2025 – Now *Mixed-Mode Ventilation in Schools: Impact on Learning, Comfort and Energy*, **Researcher**
National Research Foundation (NRF) Cities of Tomorrow R&D Programme, **S\$1,510,000**
- Oct.2025 – Now *Cost-Effective Wireless Occupant-Centric Controls for Energy-Efficient Whole Building Retrofit*, **Lead Researcher**
Building Construction Authority Green Buildings Innovation Cluster (GBIC) R&D Programme, **S\$1,256,000**
- Oct.2025 – Now *Thermoelectric cooling with novel material composites for localised cooling in naturally ventilated and mixed-mode ventilated spaces*, **Researcher**
Building Construction Authority Green Buildings Innovation Cluster (GBIC) R&D Programme, **S\$1,648,000**
- Apr.2025 – Now *AI-Driven Climate Resilient Cooling: Robust Reinforcement Learning for Mixed-Mode Ventilation*, **Lead Researcher**
Japan Science and Technology Agency (JST) and the Agency for Science, Technology and Research of Singapore (A*STAR) Joint Grant for Artificial Intelligence, **S\$345,000**
- Nov.2024 – Nov.2025 *Towards Sustainable Cooling: Occupant-Centric Mixed-Mode Ventilation Control as a Climate Adaptation Solution*, **Lead Researcher**
Korea University-National University of Singapore Strategic Partner Funds, **S\$20,000**
- Apr.2024 – Apr.2025 *Comparative Energy Analysis and Optimization of Radiant Cooling Panels in Tropical Climates*, **Participating researcher**
Gift from SANKEN SETSUBI KOGYO CO., LTD, **S\$200,000**
- Oct.2020 – May.2024 *An automatic mobile sensing platform for indoor environmental quality assessments*, **Principal Investigator**
Fund for Research & Creativity / Graduate Student Assembly Travel Fund / Scholarly Project Funding, CMU, **\$10,500**
- Oct.2022 – May.2023 *Total Estimated Cost Impact (TECI) Metrics Evaluation Project*, **Participating researcher**
United States General Services Administration, **\$300,000.**
- Oct.2022 – May.2023 *National Outdoor Air Verification (NOVA) Project*, **Participating researcher**
Fund for Research & Creativity / Graduate Student Assembly Travel Fund / Scholarly United States General Services Administration, **\$300,000.**

Honors and scholarships

2020-2023	Scholarly Project Funding × 4, Graduate Student Assembly, CMU
2020 - 2021	Letter of commendation × 2, School of Architecture, CMU
2014	Mechanical Engineering Bobcat Award, UC Merced
2013	UC Merced Graduate Summer Fellowship, UC Merced
2010	People's Scholarship, Nanjing University

Invited Talks

Mar.2026	AI in Building Controls <i>Seminar at Korea University</i>
Nov.2025	Data-driven controls for mixed-mode ventilation system via adversarial inverse reinforcement learning <i>Oral Presentation at Tsinghua Shenzhen Graduate School-NUS Workshop</i>
Jul.2025	Learning adaptive mixed-mode ventilation policies via adversarial inverse reinforcement learning <i>Poster Presentation at ICML 2025 CO-BUILD Workshop, Vancouver, BC, Canada</i>
Jun.2024	Event-based energy impact tracking and forecasting with limited measurements for rooftop units <i>Oral Presentation at ASHRAE Annual Conference 2024, Indianapolis, IN, USA</i>
Jun.2024	An expeditious spatial mean radiant temperature mapping framework using visual slam and semantic segmentation <i>Oral Presentation at CVPR 2024 CV4AEC Workshop, Seattle, WA, USA</i>
Nov.2023	Introduce Robotics to Building Management <i>Invited Seminar at Florida Agricultural & Mechanical University, Tallahassee, FL, USA</i>
Nov.2022	Improving Post-Occupancy Evaluation Engagement Using Social Robots <i>Oral Presentation at BuildSys'22, Boston, MA, USA</i>
Nov.2022	Integrated MPC-ILC Control Design for Thermal Control of a Large-Scale Variable Air Volume Reheat Systems in Buildings <i>Oral Presentation at BuildSys'22, Boston, MA, USA</i>

Service

Reviewer	Building and Environment, Energy and Buildings, Developments in the Built Environment, Journal of Building Engineering, Building Simulation (2018-Now)
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Certifications and Memberships

2016 – Present	Full Member <i>American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE)</i>
2017 – Present	Professional Mechanical Engineer (M38549) <i>Board for Professional Engineers, Land Surveyors, and Geologists - California</i>
2017 – 2021	LEED AP BD+C <i>Green Business Certification, Inc</i>
2018	Deep Learning Specialization (4LJHJDPJMNVS) <i>Coursera</i>

Technical skills

Expertise

Proficient: Building HVAC System, Building Energy Simulation, Indoor Environmental Quality, Model Predictive Control, Control Theory

Advanced: Building Automation System, Numerical Optimization, Deep Learning, Smart Grid, Reinforcement Learning, Robotics

Programming languages

Python & Pytorch, MATLAB, Julia, R, C, C++

AEC Software

EnergyPlus, IES VE, Design Builder, Modelica, Trace 700, eQuest, HOMER Pro, Autodesk Revit, AutoCAD, BlueBeam, Adobe Acrobat DC, Rhino, Grasshopper

Tools and Operating Systems

LaTeX, Git, Microsoft Office, Docker, MacOS, Ubuntu, Raspbian, Windows

Building Management System

ALC, Honeywell IQ, Siemens Apogee, Niagara Tridium, OSIsoft PI

References

Dr. Adrian Chong, FIBPSA

Associate Professor

Department of the Built Environment

College of Design and Engineering

National University of Singapore

4 Architecture Drive

Singapore 117566

Tel: +65 9821 0638

Email: adrian.chong@nus.edu.sg

Dr. Erica Cochran Hameen

Associate Professor

School of Architecture

Carnegie Mellon University

5000 Forbes Ave

Pittsburgh, PA, 15213, USA

Tel: +1 (412) 268-1905

Email: ericac@andrew.cmu.edu

Dr. Tianzhen Hong, FIBPSA, FASHRAE

Senior Scientist

Building Technology and Urban Systems

Division

Lawrence Berkeley National Lab

One Cyclotron Road

Berkeley, CA 94720, USA

Tel: +1 (510) 486-7082

Email: THong@lbl.gov

Dr. Pingbo Tang, P.E.

Associate Professor

Department of Civil and Environmental

Engineering

Carnegie Mellon University

5000 Forbes Ave

Pittsburgh, PA, 15213, USA

Tel: +1 (412) 268-8215

Email: ptang@andrew.cmu.edu

Prof. Vivian Loftness, FAIA

University Professor

School of Architecture

Carnegie Mellon University

5000 Forbes Ave

Pittsburgh, PA, 15213, USA

Tel: +1 (412) 268-1539

Email: loftness@cmu.edu

Dr. Michael Brambley, FASHRAE

Pacific Northwest National Laboratory

PO Box 999

Richland, WA 99352, USA

Tel: +1 (509) 375-6875

Email: Michael.Brambley@pnnl.gov